



1



Mark for Review



The water depth in a certain river is 78.7 feet (ft). During a 30-day period, the water depth in this river is predicted to increase by a minimum of 1 ft and a maximum of 3 ft. What is the predicted maximum water depth, in feet, during this time?

(A) 75.7

(B) 77.7

(C) 79.7

(D) 81.7

2

Mark for Review



$$5(y - 15) + 10 = 4(y - 15) + 10$$

What value of y is the solution to the given equation?

(A) 0

(B) 1

(C) 10

(D) 15

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
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- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 3.500	$3\frac{1}{2}$ 3 1/2

3

Mark for Review

If $f(x) = x + 3$ and $g(x) = 3x$, what is the value of $5f(2) - g(2)$?

Answer Preview:

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4



Mark for Review

$$(9y + 9) - (5y + 7)$$

The given expression is equivalent to $by + c$, where b and c are constants. What is the value of b ?

Answer Preview:

5



Mark for Review



What is the slope of the graph of $y = \frac{1}{2}(11x + 16) + 4x$ in the xy -plane?

(A) $\frac{11}{2}$

(B) $\frac{19}{2}$

(C) 11

(D) 15

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6



Mark for Review

A clothing store buys shirts at a wholesale price of 4.00 dollars each and resells them each at a retail price that is 280% of the wholesale price. At the end of the season, any remaining shirts are marked at a discounted price that is 80% off the retail price. What is the discounted price of each remaining shirt, in dollars?

7



Mark for Review



Which expression is equivalent to $\sqrt[12]{x^5y^5}$, where x and y are positive?

(A) $(xy)^{\frac{12}{5}}$

(B) $(xy)^{\frac{5}{12}}$

(C) $(xy)^{17}$

(D) $(xy)^{60}$

8



Mark for Review



In 6 days, a polar bear ate 25.8 pounds of fat. Which equation describes the amount of fat y , in pounds, the polar bear ate in these 6 days?

A. $y = 25.8 + 6$

B. $y = 25.8$

C. $y = \frac{25.8}{6}$

D. $y = 6$

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9



Mark for Review

If $(x + 3)^2 = 33$, what is the value of $x^2 + 6x$?

10



Mark for Review



In the xy -plane, line s passes through the point $(0, 0)$ and is parallel to the line represented by the equation $y = 25x + 5$. If line s also passes through the point $(2, d)$, what is the value of d ?

- A. 5
- B. 25
- C. 50
- D. 55



11



Mark for Review



In triangle FGH and triangle KLM , the measures of angles G and L are each 60° . The lengths of $\overline{GH}$ and $\overline{LM}$ are each 34 centimeters, and $\frac{GH}{GF} = \frac{LM}{LK}$. Which additional piece of information would be necessary to prove that triangle FGH is congruent to triangle KLM ?

- (A) The lengths of $\overline{FG}$ and $\overline{KM}$ are each 17 centimeters.
- (B) The measures of angles H and M are each 30° .
- (C) The measures of angles F and K are each 90° .
- (D) No additional information is necessary.

12



Mark for Review



A machine launches a baseball from ground level. The baseball reaches a maximum height of 92.16 meters above the ground at 2.4 seconds and hits the ground at 4.8 seconds. Which equation represents the height above ground h , in meters, of the baseball t seconds after it is launched?

(A) $h = -t^2 + 4.8$

(B) $h = -t^2 + 92.16$

(C) $h = -16(t - 2.4)^2 - 4.8$

(D) $h = -16(t - 2.4)^2 + 92.16$

13



Mark for Review



$$f(x) = 240(2)^x$$

The function f is defined by the given equation. If $g(x) = f(x - 3)$, which of the following equations defines the function g ?

(A) $g(x) = 30(2)^x$

(B) $g(x) = 80(2)^x$

(C) $g(x) = 240(8)^x$

(D) $g(x) = 240(6)^x$

14



Mark for Review



$$-3x + 6rx = 42$$

In the given equation, r is a constant. The equation has no solution. What is the value of r ?

(A) $\frac{1}{7}$

(B) $\frac{1}{2}$

(C) 2

(D) 7



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15



Mark for Review

$$\frac{x-18}{6} + \frac{y+25}{3} = -5$$

$$\frac{x-18}{3} - \frac{y+25}{6} = 5$$

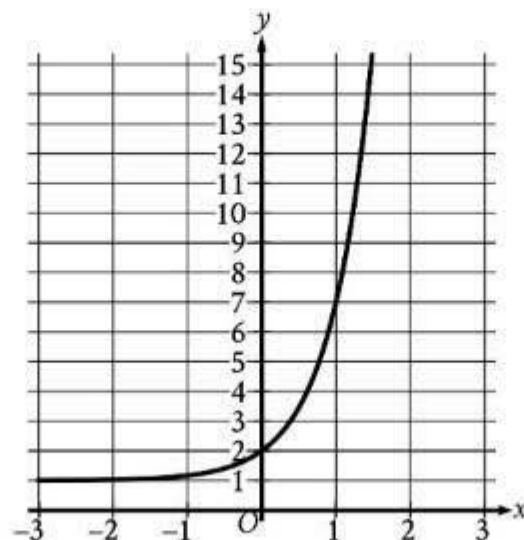
The solution to the given system of equations is (x, y) . What is the value of $x - y$?

Answer Preview:

16



Mark for Review



The graph of $y = f(x) + 1$ is shown. Which equation defines the function f ?

(A) $f(x) = 6^x$

(B) $f(x) = 2^x$

(C) $f(x) = 6^x + 2$

(D) $f(x) = 2^x + 1$



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17



Mark for Review

A raised garden is in the shape of a right rectangular prism. Its base has a width of 3 feet and a length of 27 feet, and it will be filled 24 inches high with topsoil. The total cost of the topsoil needed to fill the garden to this height is 234 dollars. What is the unit cost, in dollars per cubic yard, for the topsoil? (1 yard = 3 feet; 1 foot = 12 inches)

Answer Preview:

18

Mark for Review



The result of increasing the quantity x by 600% is 126. What is the value of x ?

(A)

18

(B)

21

(C)

474

(D)

756

19



Mark for Review



$$\frac{20}{p} = \frac{20}{q} + \frac{20}{r} + \frac{20}{s}$$

The given equation relates the positive variables p , q , r , and s . Which of the following is equivalent to q ?

(A) $p + r + s$

(B) $20(p + r + s)$

(C) $\frac{prs}{pr+ps+rs}$

(D) $\frac{prs}{20p+20r+20s}$

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20



Mark for Review

The perimeter of an equilateral triangle is 642 centimeters. The three vertices of the triangle lie on a circle. The radius of the circle is $w\sqrt{3}$ centimeters. What is the value of w ?

Answer Preview:



21



Mark for Review



$$f(x) = 3(x - a)(x - b)(x - c)$$

The function f is defined by the given equation, where a , b , and c are distinct constants. When $a < x < b$, the value of $f(x)$ is positive. The graph of $y = f(x)$ in the xy -plane contains the point (r, s) , where r and s are constants. If $s = 8$, which of the following could be true?

- I. $r < a$
- II. $b < r < c$
- III. $r > c$

(A) I only

(B) III only

(C) I and III only

(D) II and III only



22



Mark for Review



A right square pyramid has a total surface area of 36,864 square inches, and the combined surface area of the four lateral faces of this pyramid is 20,480 square inches. What is the height, in inches, of this pyramid?

- A. 48
- B. 64
- C. 80
- D. 128